Code to Connect | S&P Global & Crisil Campus Hackathon 2026

S&P100 | AI-Driven Explainable Risk Engine

An explainable AI/NLP engine for index rebalancing and portfolio stress testing — with an audit ledger behind every signal to understand and verify the system results.

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Demo Video Link:- https://youtu.be/LXFkk9zTnVs

GitHub:- https://github.com/rkrrish/IIIT_Pune-KrrishSwarnkar-CodeToConnect2026

Problem & Approach

Market-moving news — rate decisions, sanctions, earnings shocks — arrives as unstructured text across many feeds, faster than any desk can read it. Typical sentiment tools return a score nobody can trace back to its source, and stress tests are still run by hand after the event.

The Approach

- Transformer-Powered NLP Engine: Uses deep **contextual models** to deliver high-precision sentiment analysis.
- Context-Aware Impact Scoring: Integrates historical entity data to **filter market noise** and capture multi-day trends.
- Black-Litterman Re-Indexing: **Combines** market equilibrium with **subjective views** to generate stable, optimal fiduciary allocations.
- Predictive Stress Tester: Simulates macro event scenarios to quantify real-time market impact and tail risk.

Explainable design

- **Deterministic Event Ledger:** Binds every signal and weight adjustment to its unique source event ID for end-to-end auditability.
- **Explainable NLP Engine:** Eliminates black-box risks by allowing analysts to inspect inputs against outputs in real time.
- **Surgical Data Remediation:** Instantly trace, rectify, or purge faulty inputs to eliminate bad data impact across the pipeline.

System Highlights

- ProsusAI/**FinBERT model** for generating Sentiment Score ranging from -1 to 1.
- **DeBERTa-v3-base**, **fine tuned** on zeroshot/twitter-financial-news-topic (HuggingFace Dataset) for event classification (classified into 20 distinct financial topic labels)
- **Transparent and Tuneable Impact score,** allowing flexibility and tuning the system for real world deployment.
- **Modular structure** for different stages as well as for the backbone of the engine, allowing easy updates for **dealing with concept drift** and **data drift**.
- Comprehensive line graphs and bar charts generated on **a per-ticker basis**, paired with **step-by-step execution tracking** to continuously monitor intermediate outputs and **streamline analytical workflows**.

System Design & Data Flow

Figure 1.: System Pipeline

DATA INGESTION

NewsAPI
news articles

GDELT
global event data

BigData
financial news feed

Mock store · DEMO
data/mock_store_2 replayed as a time series. OR new high volume data

CORE ENGINE

ProsusAI/FinBERT
Calculates sentiment score, −1 to +1

DeBERTa-v3
Classifies events into categories, fine-tuned

Impact score calculation
1–10, deterministic formula

EWMA tracker
last 6–7 readings per ticker, to handle fake and noisy inputs

FinBERT and DeBERTa run in parallel on every item

SIGNAL

- **Composite Sentiment Score**: Aggregates exponentially weighted moving average (EWMA) and rolling sentiment scores across all tracked tickers.
- **Sentiment Momentum & Volatility**: Measures rate-of-change and standard deviation in sentiment to detect sudden risk regime shifts.
- **Event Impact Weighting**: Filters noise by scaling risk based on weighted event classification impact scores (e.g., regulatory or macro news carry higher impact weights than routine product commentary).

DOWNSTREAM MODULES

Module A · Index Rebalancer
Black-Litterman algorithm for reindexing, sentiment as views → new index weights

Module B · Stress Tester
Macro / Geopolitical event with impact > 7 → shocks → before/after P&L

FastAPI + React dashboard

- ❑ DEMO MODE: Simulates and tests system using the mock data store. Can also be used to start from an initial point, and then process huge data batch-wise to experiment or observe the changes incrementally.
- ❑ LIVE MODE: Fetches data from the source APIs in real time, and runs the pipeline
- ❑ STRESS TESTING MODE: Simulates the impact of major real-world events on a portfolio of wholesale banking assets.

UI Features and Interactions

DEMO MODE FEATURES

1. Next Timestep : Inputs the next data batch from the data source, it is processed through the whole pipeline, and changes are reflected in the dashboard.
2. Run Pipeline : Automatically inputs all the data from the data source batch wise and changes are reflected in the dashboard
3. Inspection of Audit ledgers to check which input caused a specific output.
4. Graph visualization to observe the change in market and indices.

LIVE MODE FEATURES

- Fetches fresh articles for the 20 index tickers from NewsAPI, GDELT and BigData and runs the same pipeline end to end. Each refresh fetches a limited set of articles.
- Portfolio's value before and after the stress test is compared using graphs, highlighting the impact of the event.

Audit ledger
Allows verification any result and action taken by the system against the input that produced it.

Key Results

92.98%

DeBERTa-v3 **event-classifier accuracy** (F1 0.927 vs FinBERT 0.897)

Simulation

re-weighted in one live run; every change logged with its event ID, weights still sum to 100%

Easy to add sources

NewsAPI, GDELT and BigData all appear in the live-run ledger. Modular structures allows to integrate various sources easily for data ingestion into the system pipeline

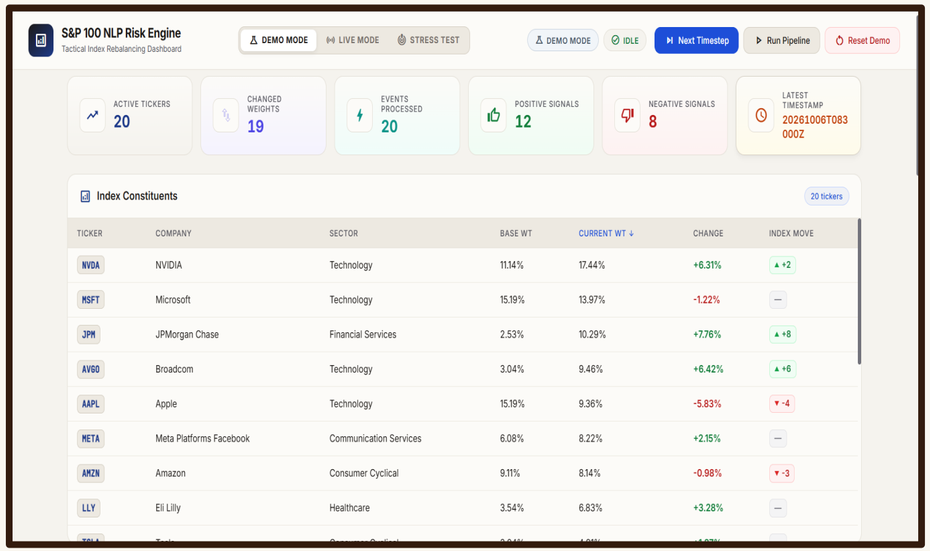


Figure 2.: Dashboard result of running the full pipeline.

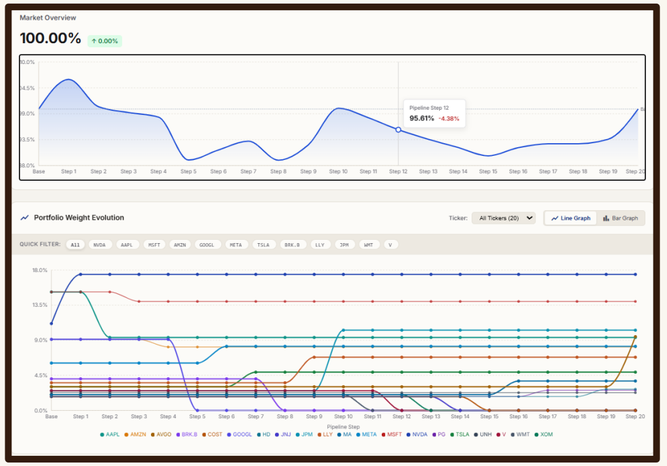


Figure 3.(a)



Figure 3.(b)

Figure 3. (a) and (b): Visualization of results of the full pipeline execution.

Efficiency	Traditional/Manual method	Proposed system
Reacting to news	One headline moves the weight	Exponentially Weighted Moving Average over 6–7 readings smooths single-post noise
Weighting method	Mean-variance: concentrated corner solutions	Black-Litterman, anchored to the benchmark
Stress testing	Manual scenario run after the event	Event class + impact > 7 maps to a shock scenario and position-level P&L

Table 1: Comparison of system architecture with Traditional methods

Why It Matters

Asset managers & index providers

Sentiment-aware tilts that are smoothed over several readings, anchored to a benchmark and traceable, so weights do not swing on a single headline.

Wholesale-banking risk desks

A high-impact Macro or Geopolitical event (impact > 7) maps directly to a calibrated shock scenario and a position-level P&L, giving a first view in seconds of what a desk would otherwise assemble by hand.

Compliance & model-risk teams

The ledger answers “why did this weight change?” with the event ID, rolling sentiment, impact score and old and new weight, ready for audit and review.

Builders & researchers

A modular stack on free and open sources with a Docker one-command run; the engine can feed any new downstream module.

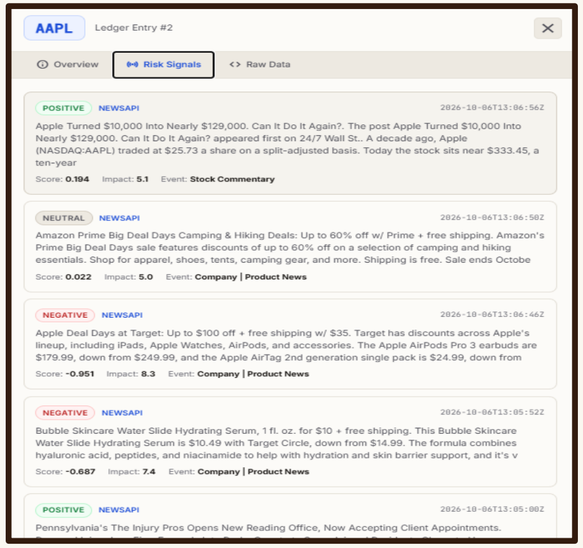


Figure 6. (a)

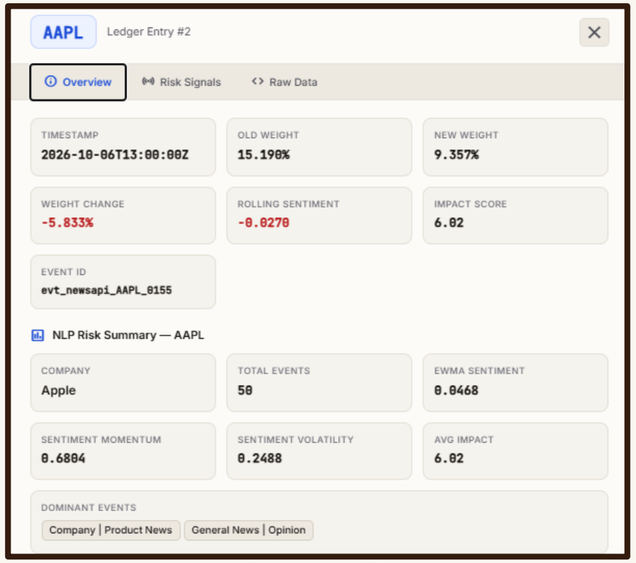


Figure 6. (b)

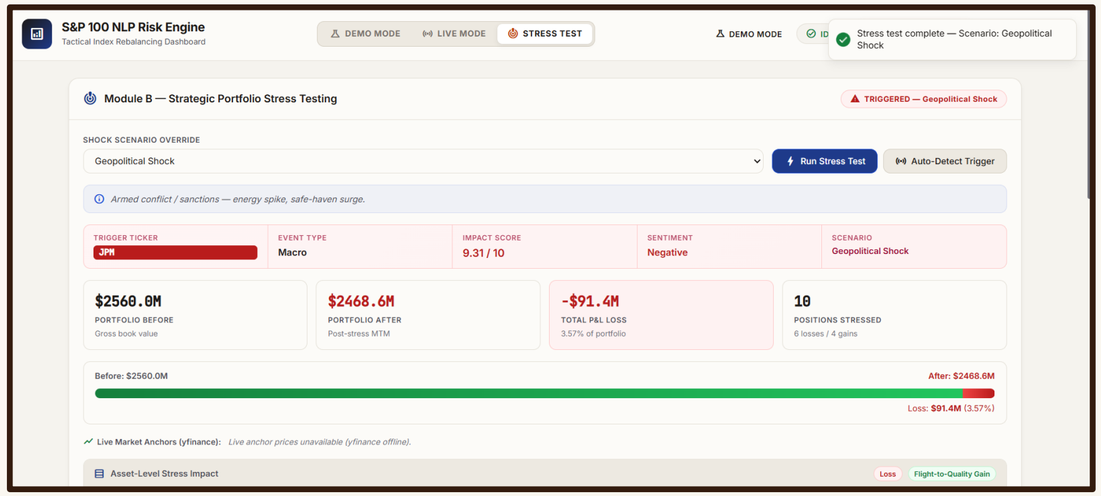


Figure 4.: Stress testing module results

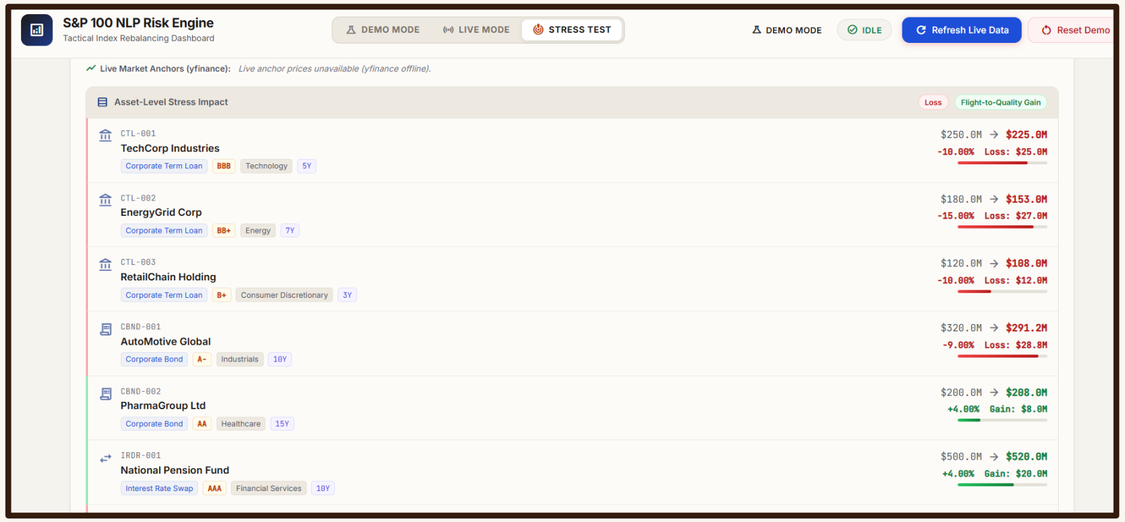


Figure 5.: Stress Testing module’s tabular comparison

Figure 6. (a) and (b): Inspection of ledger entries, allowing to see outputs of different stages of the pipeline to verify the final results and changes.

Limitations & Next Steps

Assumptions & gaps

- **Synthetic & External Input Data:** Demo mode leverages historical date ranges from GDELT and NewsAPI to feed simulation inputs, while portfolio assets remain synthetic **Randomized Initial Parameters:** Initial weights and baseline indices are randomly generated rather than derived from real-world baselines, which can cause non-realistic initial jumps during simulation.
- **API Constraints & Reliability:** Reliance on free-tier APIs introduces rate limits and latency bottlenecks, occasionally leading to delayed responses or timeouts that disrupt pipeline execution.
- **Lack Evaluation for Pipeline as a whole system:** Current architecture lacks real-world comparison, hence no evaluation metrics for the output of the system, we can only evaluate the output of NLP engine using existing labelled datasets.
- **Heuristic Weighting:** Impact and event weights are manually set $(0.40/0.35/0.15/0.10;\ 0.35-0.95)$ rather than learned or back-tested.
- **Event Classifier Scope:** The event classifier is trained primarily on tweet-style headlines and can experience classification drift when processing longer, unstructured news articles.
- **Live Ingestion Throttling:** Live mode fetches a limited volume of articles per refresh cycle.

Future work

- **Data-Driven Weight Calibration:** Replace hand-set parameters with machine learning models (e.g., regression or neural network classifiers) trained on labelled historical outcomes to optimize impact and event weighting.
- **Input Correction Mechanisms:** Build administrative overrides or feedback loops to correct, purge, or mask previously ingested erroneous data, preventing downstream manipulation or unrealistic non-truth results.
- **Expanded Ingestion & Fallbacks:** Scale up data sources to include targeted social media feeds alongside GDELT and NewsAPI, supported by API rate-limit management and caching fallbacks to mitigate free-tier latency bottlenecks.
- **Long-Form Context Handling:** Integrate chunking and summarization layers for the NLP event classifier to maintain accuracy when processing long-form financial articles instead of short-form tweets.
- **Live Market Syncing:** Implement continuous syncing with recent market values to align the analysis pipeline with current macro conditions, replacing randomized initializations with live index benchmarks in demo mode.
- **Empirical Covariance Estimation:** Transition from synthetic single-factor models to real-time covariance matrices estimated directly from empirical asset price history.
- **Extended Market Data Pipelines:** Expand yfinance integration beyond basic stress testing to anchor pricing and validation across all analytical modules.

THANK YOU!